THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 12: **Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of the data 4, 7, 9, 10, 15, 21 is: 1

(a) 9 **(b)** 11 **(c)** 10.5 **(d)** 13

2. The mean of the data 5, 6, 6, 9, 11, 13, 14 is: 1

(a) 9.14286 **(b)** 11.1429 **(c)** 7.14286 **(d)** 10

3. The median of 8, 8, 10, 12, 14, 18 is: 1

(a) 11 **(b)** 10 **(c)** 14.6667 **(d)** 12

4. The median of 45, 52, 38, 60, 55, 40 is: 1

(a) 51.3333 **(b)** 47.5 **(c)** 49.5 **(d)** 48.5

5. The mean of 4 numbers is 11. 3 of them are 8, 10, 14. The remaining number is: 1

(a) 10 **(b)** 12 **(c)** 14 **(d)** 17

6. In a dot plot, each dot represents: 1

(a) one observation **(b)** the median **(c)** the mean **(d)** ten observations

7. The mean is called the 'balancing point' because: 1

(a) the total deviation on both sides is equal
(b) it is the middle value
(c) it is the most frequent
(d) it is the largest

8. The average of 4, 6 and 8 is: 1

(a) 6 **(b)** 7 **(c)** 5 **(d)** 8

9. The median of an even number of observations is: 1

(a) the first value **(b)** the larger middle value
(c) mean of the two middle values **(d)** the mean

10. A line graph is drawn by joining plotted points with: 1

(a) line segments **(b)** circles **(c)** sectors **(d)** bars

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. The mean of 5 numbers is 20. If one number 28 is removed, what is the mean of the rest? 2

12. Why is the median preferred to the mean for incomes? 2

13. State when a line graph is suitable. 2

14. Find the range of 14, 8, 22, 5, 19. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. The data 100, 120, 140, 160, 180 is given. Compute the mean and median and comment on which is a better measure. 3

16. The data 45, 52, 38, 60, 55, 40 is given. Compute the mean and median and comment on which is a better measure. 3

17. The data 3, 9, 6, 12, 15, 21, 18 is given. Compute the mean and median and comment on which is a better measure. 3

OR

The data 16, 18, 20, 22, 24, 54 is given. Compute the mean and median and comment on which is a better measure. 3

18. The mean of 5 observations is 16. If 4 of them are 12, 15, 18, 20, find the remaining observation. 3

OR

The mean of 4 observations is 11. If 3 of them are 8, 10, 14, find the remaining observation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** Five friends' pocket money (₹): 200, 250, 300, 350, 900. 5

(i) Find the mean.
(ii) Find the median.
(iii) Which better represents them?
(iv) What happens without 900?

OR

(i) Show that the mean of the first n natural numbers is $(n+1)/2$. 5
(ii) Hence find the mean of the first 50.

20. (i) A class of 30 has mean 62. A class of 20 has mean 72. Find the combined mean. 5
(ii) Compare.

OR

Case study: The monthly rainfall (cm) in a town is Jan 2, Feb 3, Mar 5, Apr 8, May 12, Jun 20. 5

(i) Draw (describe) the line graph.
(ii) Find the mean.
(iii) Between which months is the rise greatest?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND